Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 17 – At-risk users



Content warning: general mentions of sexual harassment, violence, and abuse

Last time

- Responders' view of IPV
- Tech abuse clinics
  - Strengths
  - Limitations
- Trauma-informed computing and applications
  - Mitigating researcher trauma
  - Evidence collection

Today

- At-risk users
  - Introduce a framework for understanding contextual risk factors, protective strategies, and challenges
- Introduce a new threat model: locally-powerful adversaries
  - Example of project building on at-risk SoK to identify this new threat model

At-risk users

An *at-risk user* is anyone with risk factors that increase their chances of being attacked digitally and/or suffering disproportionate harms from an attack

- Journalists and political activists
- Children
- Older adults
- Individuals with disabilities
- Abuse survivors
- Vulnerable workers
  - Sex workers
  - Gig workers

What could make a user at-risk

- Security: adversary with capabilities + motives
- Human-computer interaction: platform design choices

Another example: gig work

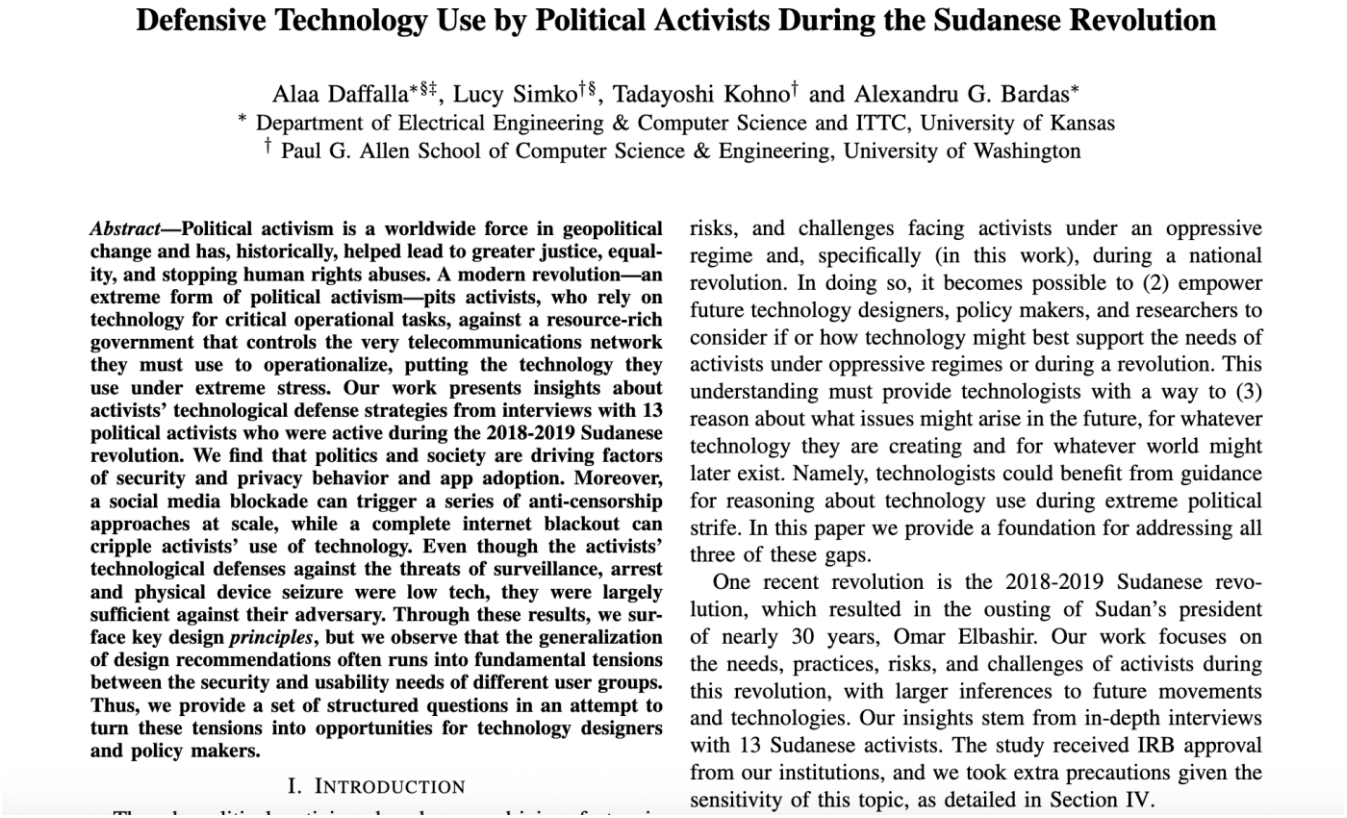
- Algorithmic systems assign work based on ratings
- Higher ratings = more and better-paying work
- This limits workers' agency, which can be harmful in unsafe situations
- Example: panic button



One group from reading: Political activists

Adversary might:

- Control the network
  - Influence over telecoms and ISPs
  - Can surveil SMS and phone calls
- Censor and shut down access
  - Block popular VPNs by IP range
  - Block certain platforms
  - Full internet shut down
- Spread misinformation
- Arrest people & seize devices
- Infiltrate online groups
  - Can join activist groups and chats, since membership constantly changes



Defensive Technology Use by Political Activists During the Sudanese Revolution

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***Abstract*—Political activism is a worldwide force in geopolitical change and has, historically, helped lead to greater justice, equality, and stopping human rights abuses. A modern revolution—an extreme form of political activism—pits activists, who rely on technology for critical operational tasks, against a resource-rich government that controls the very telecommunications network they must use to operationalize, putting the technology they use under extreme stress. Our work presents insights about activists' technological defense strategies from interviews with 13 political activists who were active during the 2018-2019 Sudanese revolution. We find that politics and society are driving factors of security and privacy behavior and app adoption. Moreover, a social media blockade can trigger a series of anti-censorship approaches at scale, while a complete internet blackout can cripple activists' use of technology. Even though the activists' technological defenses against the threats of surveillance, arrest and physical device seizure were low tech, they were largely sufficient against their adversary. Through these results, we surface key design *principles*, but we observe that the generalization of design recommendations often runs into fundamental tensions between the security and usability needs of different user groups. Thus, we provide a set of structured questions in an attempt to turn these tensions into opportunities for technology designers and policy makers.**

I. Introduction

risks, and challenges facing activists under an oppressive regime and, specifically (in this work), during a national revolution. In doing so, it becomes possible to (2) empower future technology designers, policy makers, and researchers to consider if or how technology might best support the needs of activists under oppressive regimes or during a revolution. This understanding must provide technologists with a way to (3) reason about what issues might arise in the future, for whatever technology they are creating and for whatever world might later exist. Namely, technologists could benefit from guidance for reasoning about technology use during extreme political strife. In this paper we provide a foundation for addressing all three of these gaps.

One recent revolution is the 2018-2019 Sudanese revolution, which resulted in the ousting of Sudan's president of nearly 30 years, Omar Elbashir. Our work focuses on the needs, practices, risks, and challenges of activists during this revolution, with larger inferences to future movements and technologies. Our insights stem from in-depth interviews with 13 Sudanese activists. The study received IRB approval from our institutions, and we took extra precautions given the sensitivity of this topic, as detailed in Section IV.

Goal: Safer informal abuse reporting for labor abuse

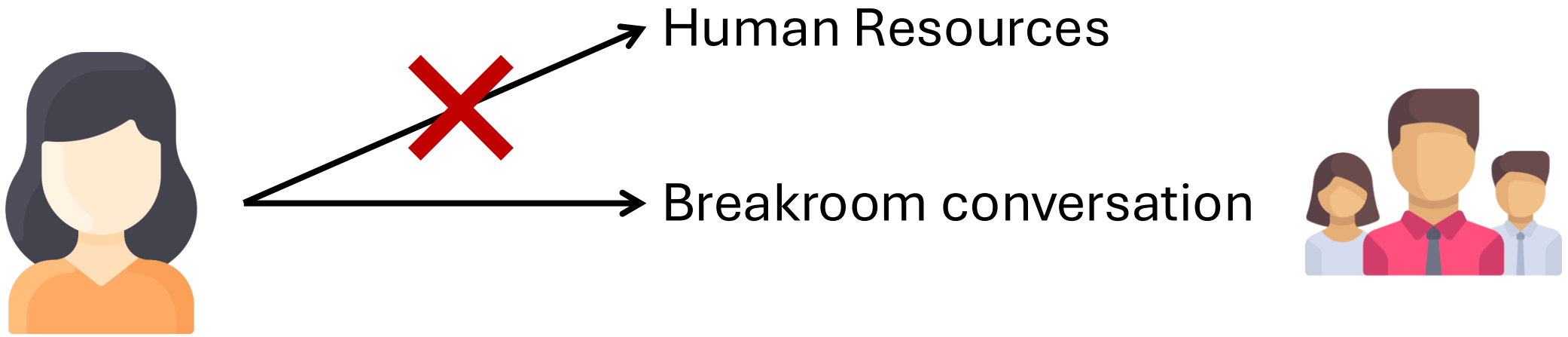


Rivera et al., Goals, Risks, and Safety Practices in Online Labor Abuse Disclosures, SOUPS 2026

Labor abuse includes:

- Economic exploitation
  - Wage theft
  - Non-payment
  - Breach of contract
  - Financial scam
- Discrimination
- Health and safety violations resulting in injury or illness
- Sexual misconduct
  - Harassment
  - Unwanted touching

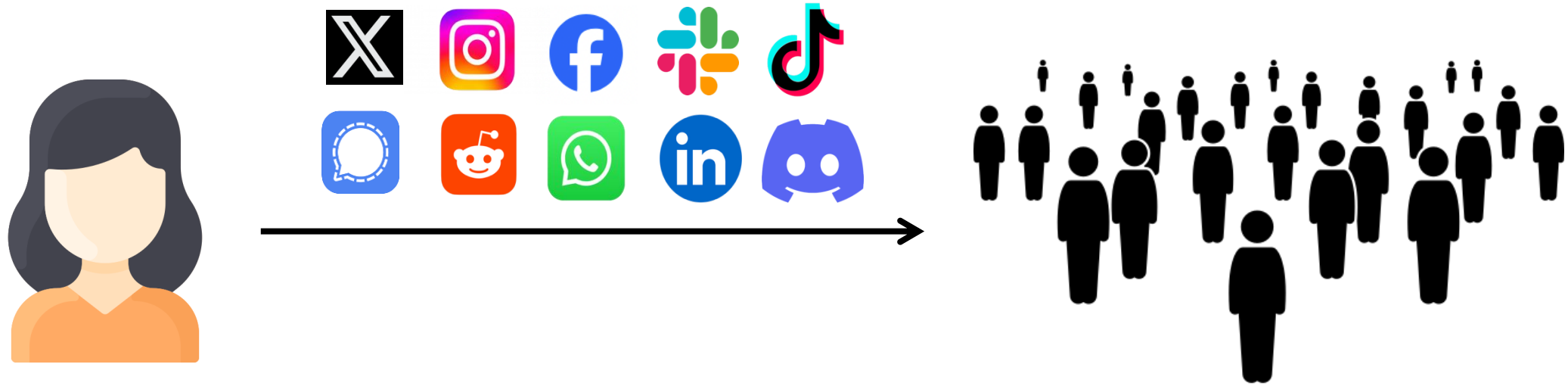
Goal: Safer informal abuse reporting for labor abuse



**whisper network: an informal chain of information passed privately between people*

*Johnson, Carrie Ann. "The purpose of whisper networks: a new lens for studying informal communication channels in organizations." *Frontiers in Communication* 8 (2023): 1089335

Labor abuse survivors seek targeted visibility in *digital whisper networks*

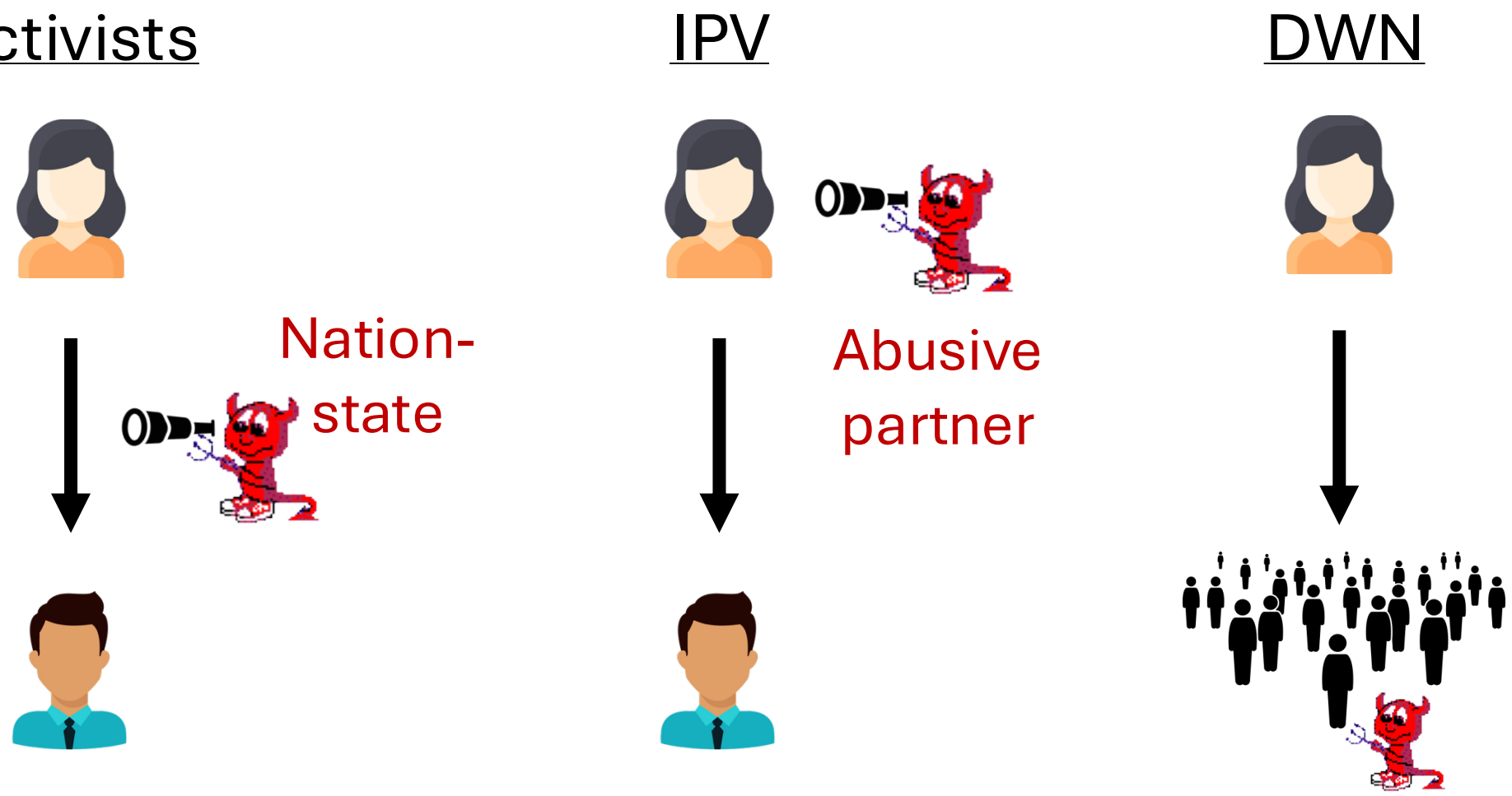


Targeted visibility: to be widely believed and acted on by supporters, while staying hidden from adversaries.

Targeted visibility is a contextual risk factor

- Prominence (Warford et. al.): Individuals are targeted for their notability in a population
- Prominence is static – inherent attribute of an individual or a population
- Targeted visibility is dynamic & tied to information flow
  - Safety calculus (disclosure goals vs risks)
  - Audience construction *across* platforms

A new threat model: locally-powerful adversaries



Research questions

Structure: What types of technologies do participants use to disclose their experiences?

Goals: What goals do participants have when sharing their experiences? How do these shape which technologies they use?

Risks: What risks do participants experience or perceive when using DWNs? How do these relate to the disclosure and/or the tech used?

Mitigations: What decisions do participants make during the process of disclosure? How do they intersect with disclosure goals?

Participants' have three goals for sharing in DWNs

Solicit support

"**I was looking for, has anybody else had these experiences, like am I crazy?** Just kind of looking for validation". (P16)

Broadcast experience

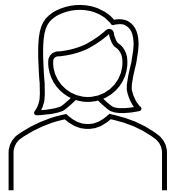
"I was super connected with a lot of other models and photographers online. **If people were to work with this person in the future, at least they would be aware**". (P15)

Organize community

"[I was looking to make] the department and organization **a better and safer place for all of us**". (P9)

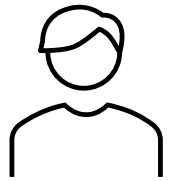
Participants face risk from platforms & goals

- Uncontrolled re-sharing
- Retaliation
- Reputational damage
- Harassment

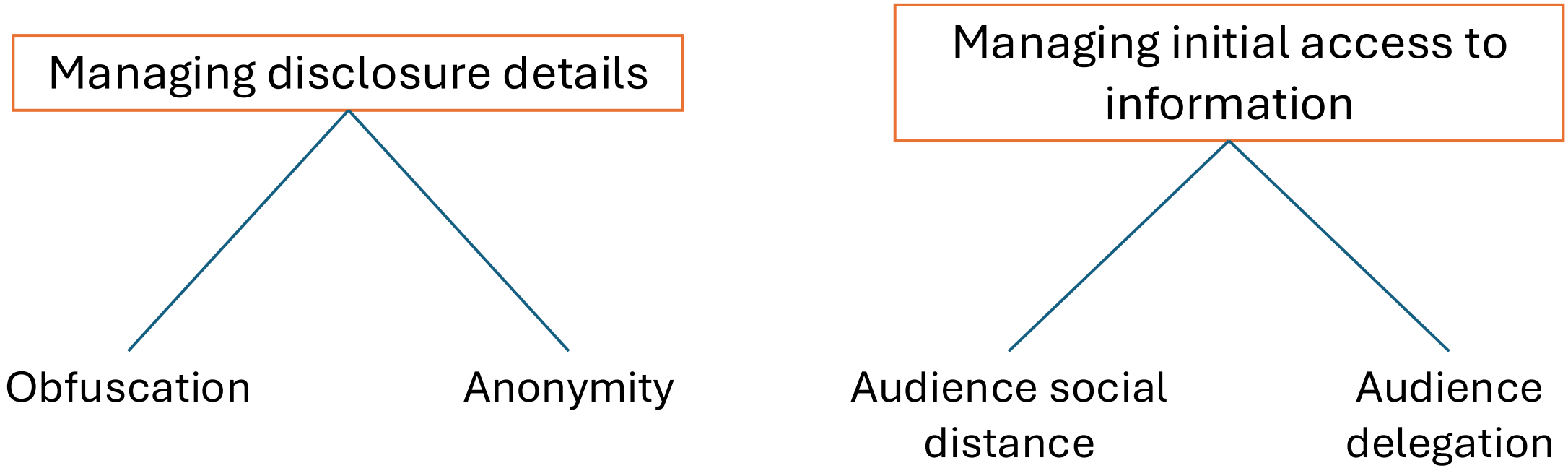


"**They might take the whole text, screenshots, and everything**, and send them back to whoever you had a disagreement with." (P10)

"Someone said I was lying, that the **admins should take me out of the group**, and that maybe I am the scammer". (P4)



Participants take two types of mitigative steps

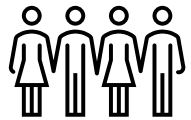


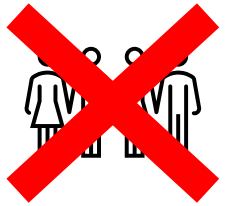
Audience management: distance & delegation

Participants sought an audience at the "Goldilocks" distance:



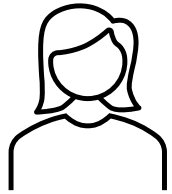
close contacts





no context

Many delegate to **group** admission policies:



"you have to drop your name, your number, your home address. You have to submit a 360 degree picture to show you're a real person. [And] a video introducing yourself and telling them what you do." (P4)

Protective behaviors

At-risk users employ **protective behaviors, or practices** that they *believe* will help them prevent, mitigate, or respond to digital safety risks

- Often a response to institutional or platform protective mechanisms being insufficient
- Remember: groups and communities are experts in their experiences
  - We can learn from their protective behaviors!
- On Monday: protective behaviors in digital intimacy, labor, and the intersection of these

Reminder: AI use

DO NOT use AI to write your written reports for this class (or to produce any other deliverables)